Romina Saljooghian

Robotics AI & Machine Learning Engineer

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Profile

MSc graduate in Automation and Control Engineering with experience in deep learning, computer vision, agentic AI, and control systems. Built end-to-end ML pipelines from foundation-model orchestration to production REST APIs with LangGraph. Seeking roles in ML Engineering, CV, or Robotics AI.

Education

MSc · Automation and Control Engineering Sep 2022 – Mar 2026
Politecnico di Milano Milan, Italy

- **GPA** 107 / 110 — **Thesis:** *Semantic Characterization of Robotic Environments*
- Open-vocabulary scene graph generation for robotic manipulation using foundation models (CLIP, SAM, Grounding DINO, RAM) — MERLIN Lab.
- **Selective Coursework:** Neural Networks & Deep Learning · Control of Industrial Robots · Multiagent Systems · Software Engineering · Model Identification · Data-Driven Control · Autonomous Vehicles

BSc · Electrical Engineering (Minor: Control) Sep 2016 – Aug 2020
K. N. Toosi University of Technology Tehran, Iran

Projects & Research

Scene Graph Generation for Robotic Manipulation — Python . PyTorch . CLIP . SAM . GPT API . Hydra 2025 – 2026

- Designed a modular open-vocabulary pipeline chaining 4 foundation models; **6× improvement** over end-to-end baseline (PredCls mR@100: 48.90%)
- Built VLM-based annotation pipeline (GPT + Gemini) with structured prompts, producing 246 spatial-relationship triplets for a novel robotics dataset
- Zero-shot cross-domain transfer to manipulation scenes (SGDet mR@20: 16.13%) with no fine-tuning

Visual QA Agent — Python · FastAPI · LangGraph · Anthropic API 2026

- Built agentic backend answering visual questions about images and PDFs through a **4-node LangGraph pipeline** with conditional error routing at every stage
- Dual-provider VLM abstraction (Claude + GPT-4o); PDF-to-image conversion via PyMuPDF; 11 automated tests with full coverage

Plant Leaf Disease Classification — Python · TensorFlow · FastAPI · MLflow · pytest 2024

- 5-model ensemble CNN achieving **86% accuracy**; async REST API with Grad-CAM explainability endpoint
- MLflow experiment tracking with nested runs; 6-module pytest suite covering API, data pipeline, and model registry

Time-Series Forecasting — Python · TensorFlow · Keras 2024

- Benchmarked 6+ architectures on 48K time series; Conv. LSTM with autoregressive decoding achieved **test MSE of 0.01**

CNC Machine — 2-DOF Pantograph Robot — MATLAB · Simulink . Quanser SRV02 Hardware 2024

- Deployed 9 controllers (PID, LQR, MPC, Kalman) on real hardware; MPC: **0.26 s settling, <0.5% overshoot**
- Near-perfect 2D trajectory tracking with KF + LQR; validated in both time and frequency domains on physical plant

Networked Control of a Vehicle Platoon — MATLAB 2025

- LMI-based state-feedback controllers for a 4-vehicle platoon under 6 communication topologies; benchmarked centralized vs. distributed cost and convergence trade-offs

Work Experience

Delphi Software Developer Jan 2020 – Sep 2021
Torfeh Negar Holding Tehran, Iran

- Built production UI and backend modules for Holoo accounting software serving thousands of users
- Optimized SQL Server queries; resolved bottlenecks and stabilized nightly batch processes

Instrumentation & Control Intern Jun 2019 – Sep 2019
Tavan Gostare Zino Tehran, Iran

- Designed and tested industrial grounding and earthing systems for IEC standards compliance

Skills & Certifications

Languages	English (Fluent) Italian (A2) Persian (Native)
Programming	Python (*freeCodeCamp*) MATLAB SQL (*freeCodeCamp*)
ML & Deep Learning	PyTorch TensorFlow Keras scikit-learn OpenCV Hugging Face
LLM & Agentic AI	LangGraph (*LangChain Academy*) LangChain OpenAI API Anthropic API
Backend & API	FastAPI REST API Pydantic pytest
MLOps & Tools	MLflow Hydra Git Linux LaTeX Automation Explorer (*UiPath*)
Control & Robotics	Simulink MPC LQR Kalman Filter PID Kinematics System ID